KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2019
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

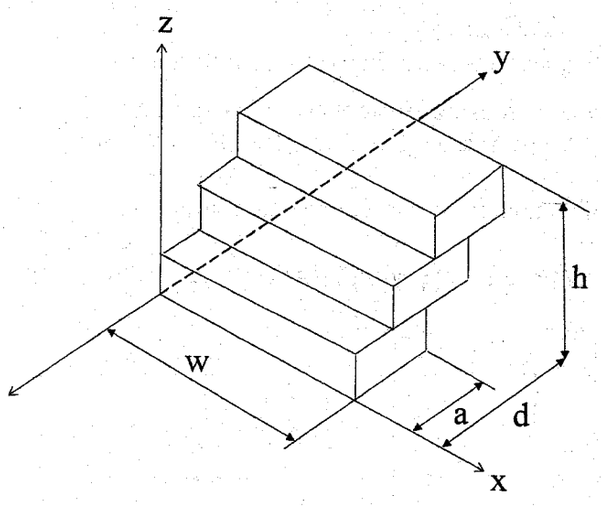
1. a) Define and mention the importance of following transformation in a Computer graphics system. (10)
   i) Modelling transform.
   ii) Viewing transform.

   b) In 2D, it is possible to rotate data using a sequence of a shear in x, followed by a shear in y, and then a non-uniform scaling transform. Rather than using the rotation matrix, you are instructed to use the above mentioned method to perform a θ° rotation. In that case- (15)
   i) What should the X and Y scale factor be?
   ii) Determine the X and Y shear terms and their relationship. As a reminder, the homogeneous shear matrices are given below.

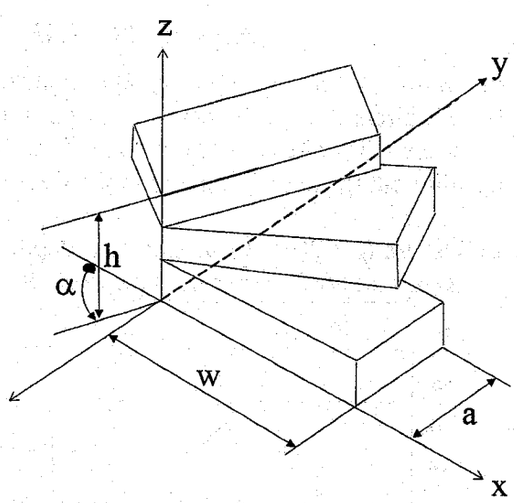
$$shear(x) = \begin{bmatrix} 1 & Sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad shear(y) = \begin{bmatrix} 1 & 0 & 0 \\ Sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

   c) Let a triangle is defined by the points A (1, 0, 0), B (0, 2, 0) and C (0, 1, 1). Determine if a ray with $\mathbf{s} = i + j + 3k$ and $\mathbf{d} = -2k$ intersects the triangle or not? First find the intersection of ray and triangular plane and then use Badouel's algorithm to verify that the intersection lies within the triangle edges. (10)

2. a) Suppose you have to draw a stair having N steps, each step has width w and depth a; the total stair height is h; for a straight forward stair, the total depth is d. See the left figure below, where a stair with N = 3 is shown. Assume that a function *Drawstep(M)* is available to draw a step, subject to a homogeneous transformation M. A step in local coordinates is a cube $-1 \le x,y,z \le 1$, centered around the origin. Furthermore, suppose that a standard set of basic modeling transformation functions for translation, rotation and scaling are available. (11)
   i) Define a matrix M_i using basic transformation functions along with their parameters to draw step i, i =1, …, N for the straight forward stair.
   ii) Consider a spiral stair (see the right Figure below), where the total rotation is α. Define the matrix M_i again for drawing a step i, i =1, …, N for this spiral stair.



Straight Forward Stair



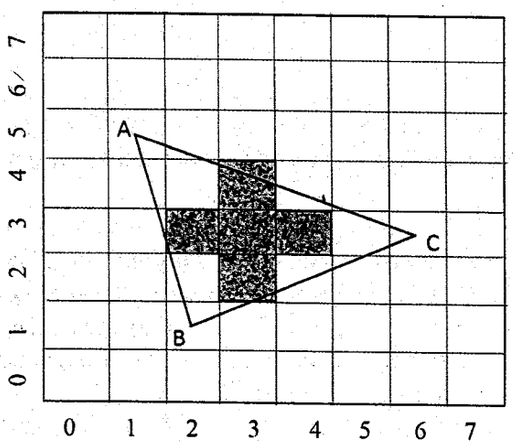
Spiral Stair

   b) Find mirror reflection of a point P (2, 2, 1) about a plane defined by the points (1, 0, 1), (1, 0, 0), and (0, 1, 0). (12)

c) Derive a transformation matrix that transform a world coordinate point to camera coordinate. Be sure to explain how the rotation matrix involved in above transformation is derived. Also consider the camera is defined by the three parameters *position, look-at* and *up-vector.*

3. a) Compare between perspective and axonometric projection. (05)
   b) Derive a projective transformation matrix that perspectively project any point in view-coordinate system (VCS) onto a projection plane parallel to xy-plane and placed at distance –d. Consider the center-of-projection (COP) is at C (x_c, y_c, z_c). (12)
   c) A triangular plane defined by the points (0, 2, 0), (3, 3, 0) and (1, 0, 1) is illuminated by a monochromatic point light source with light intensity 0.9 and positioned at (4, 6, 2). Find the vertex intensities using Phong illumination model if the viewer is at (1, 1, 5), diffuse and specular reflectivity co-efficient of the surface are 0.6 and 0.8 respectively. Also, consider the shininess factor = 8, no ambient light, radial attenuation constraints $a_0 = 0$, $a_1 = 1$, $a_2 = 0$, and vertex normal equals to the surface normal. (15)
   d) State the necessity of near clipping plane while defining a view volume. (03)

4. a) Explain how the Ray tracing method integrates the process of projection, scan-conversion and shadow generation. (10)
   b) Let a triangular face is rasterized to pixel coordinates A (1, 5), B (2,1), C (6,3). Say, RGB vertex intensities are I_A (0.9, 0.1, 0.2), I_B (0.1, 0.8, 0.2) and I_c (0.2, 0.3, 0.7). Use Gauraud shading method to render the triangular face showing the interval pixel intensities of the highlighted pixels shown in figure below. (14)



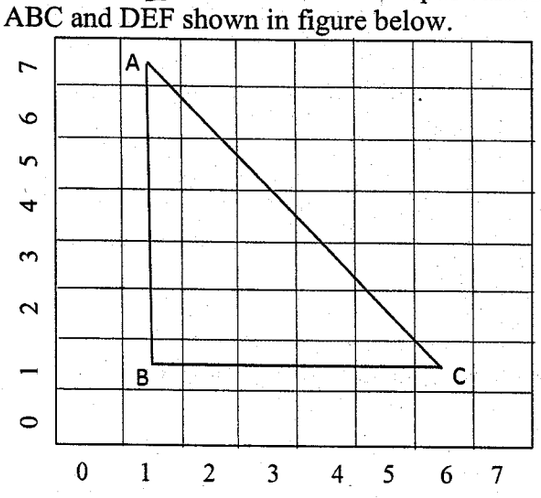
   c) Formulate a transformation matrix that converts an asymmetric parallelepiped view volume to a canonical view volume. Also mention, why should we do the conversion? (11)

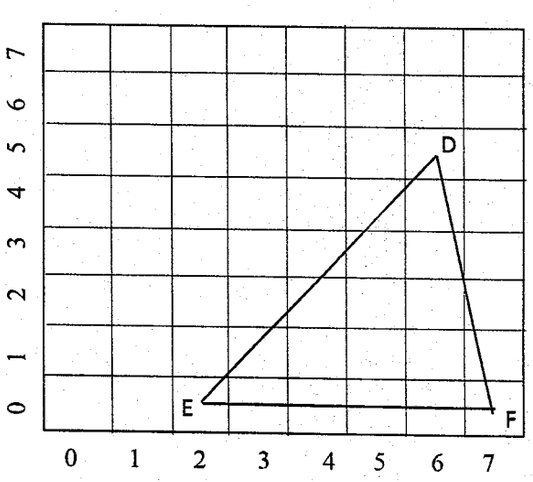
SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Digitize a line (3, 12) to (6, 2) on a raster screen using Bresenham's straight line algorithm. (10)
   b) If a circle's origin is centered at point (30, 40) and its radius is equivalent to 8, then rasterize the circle with midpoint circle algorithm. (10)
   c) Find out the decision parameter values of a quadrant of an ellipse using midpoint ellipse algorithm. (15)

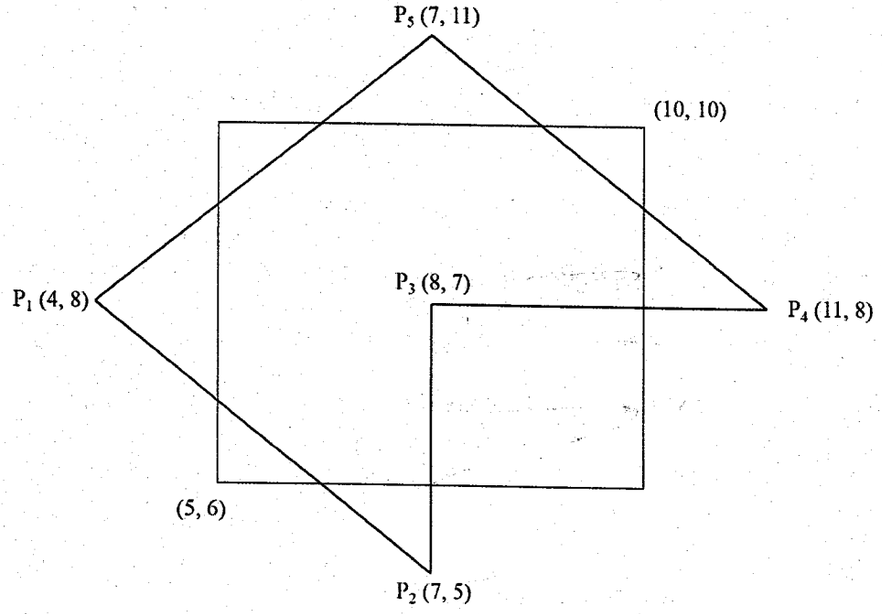
6. a) Use an 8x8 Z-buffer to store the depth values after applying Z-buffer algorithm onto the triangles ABC and DEF shown in figure below. (17)





Here the z-coordinates of the points A, B, C, D, E and F are 5, 5, 5, 2, 3 and 8 respectively. Find the internal Z-values of the triangle before applying the algorithm and show Z-buffer contents after each step. Also show the frame buffer content if the triangle surface of ABC and DEF are RED and GREEN color respectively.

b) Consider the polygon defined by the points P_1, P_2, P_3, P_4 and P_5 as shown in figure below. Use Sutherland-Hodgeman algorithm to clip polygon against the rectangular window defined by the point $(x_{min}, y_{min}) = (5, 6)$ and $(x_{max}, y_{max}) = (10,10)$. Compute the final points list. (13)



c) Define fractal. Draw the first three iterations of Koch snowflake fractal. Show each iteration separately. (05)

7. a) Write about the differences between Object-space method and Image-space method. (08)
   b) How painting priority of a surface is determined by painter's algorithm? (10)
   c) What are the major mapping methods? Give description of them. (07)
   d) Give description of Linear Texture and Cylinder Texture mapping in 2d. (10)

8. a) What is aliasing problem? How one can solve it? (10)
   b) Give an example of special case in scan-line polygon filling algorithm. (05)
   c) What do you know about geometric continuity and parametric continuity? Explain C^0, C^1, C^2 (10)
   d) Calculate Hermite Blending Functions from cubic Hermite Spline Curve having four control points. (10)